Supplementary Step 4 Visualization Report

Overview

Metric	Value
Representative geometry session	PID001 / 2025-11-06_008
Included participant-sessions from Step 4	36
Abstract onset-locked epochs used	539
Concrete onset-locked epochs used	536
Trials dropped for fixed epoch window	0
Epoch window	-3.5 s to 13.2 s around question onset
Baseline	-2.0 s to 0.0 s
Single-time comparison	9.0 s

Rendering note. The original example requested a brain-surface sensor rendering. In this environment, the report uses the participant digitization point cloud together with source, detector, channel, and pair geometry because the local `pyvista`/`fsaverage` rendering stack is not available.

Representative Sensor Geometry

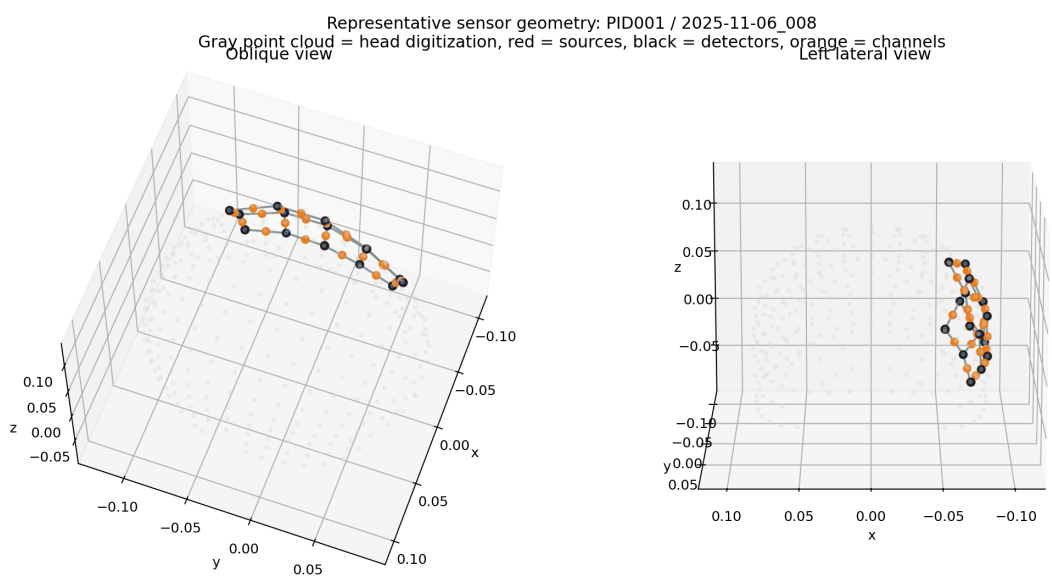

Figure 1: Representative optode, channel, and source-detector-pair geometry for one included participant-session. Short pairs are shown with dashed orange lines.

Condition-Specific Topography Over Time

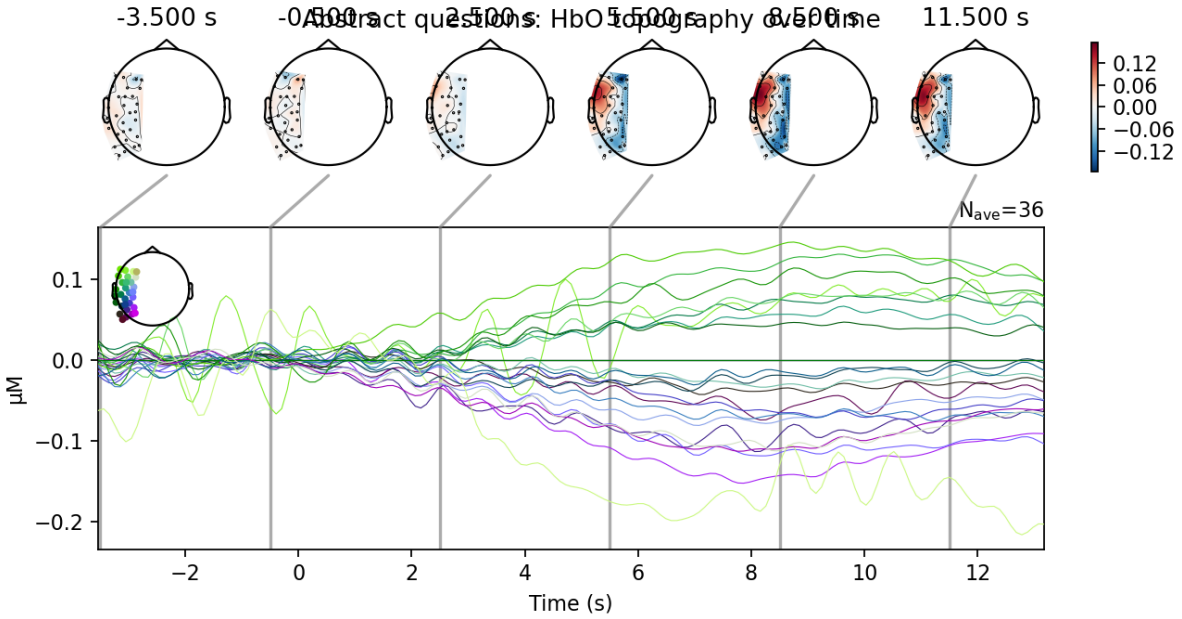

Figure 2: Grand-average HbO topography over time for abstract questions, aligned to question onset.

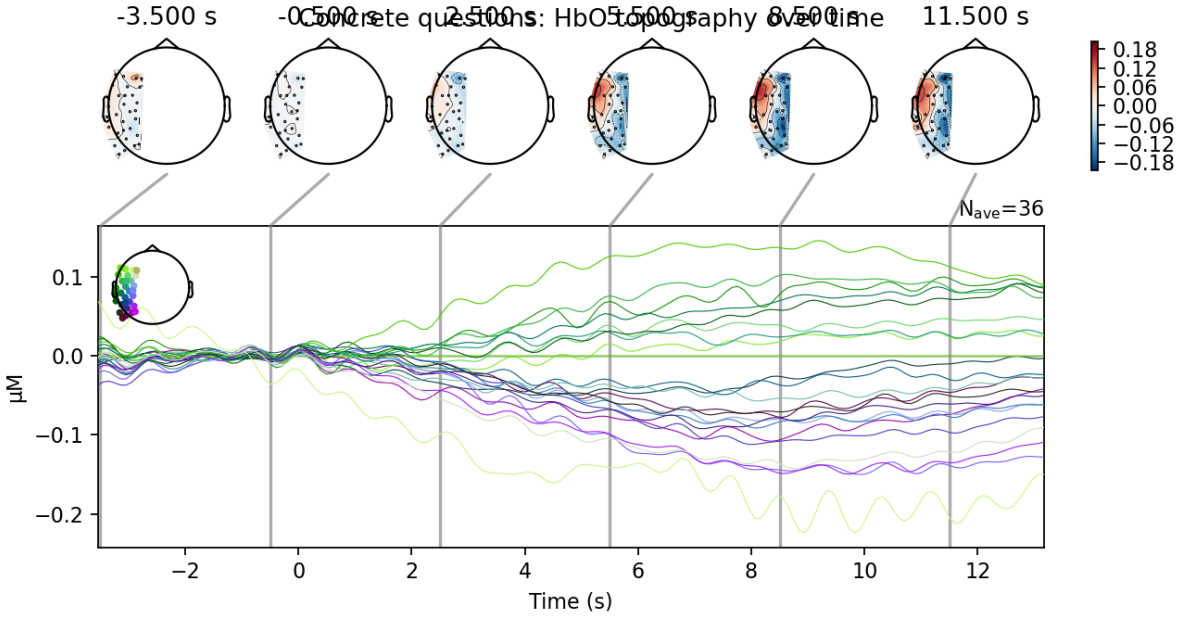

Figure 3: Grand-average HbO topography over time for concrete questions, aligned to question onset.

Single-Time-Point Condition Comparison

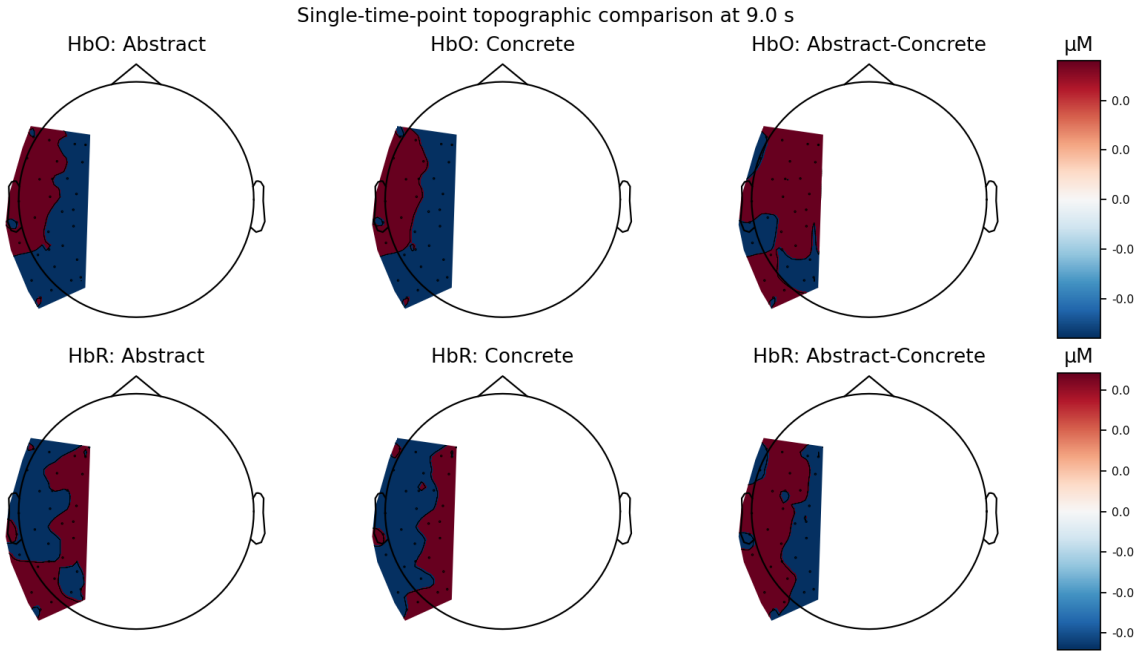

Figure 4: Condition comparison at one time point for HbO and HbR, with the abstract-minus-concrete difference shown in the rightmost column.